Yingte Xu

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Research Summary

My research applies formal methods to study programming theory for quantum computing, including semantics, verification techniques, and proof automation. The aim is a practical toolchain that can naturally encode and verify quantum algorithms and protocols. Recent work also explores machine learning and LLM-based coding agents as a means toward this automation.

Education & Employment

Ruhr University Bochum / MPI-SP — **Bochum, Germany**
PhD Student & Research Assistant — *2024–2026 (expected)*
Advisor: Gilles Barthe
Thesis: Automation of Formal Languages for Quantum Computing

Institute of Software, Chinese Academy of Sciences — **Beijing, China**
Internship — *2022–2023*

Lanzhou University — **Lanzhou, China**
Bachelor of Science (Physics) — *2018–2022*

Honors & Awards

Winner of CHES 2025 Deep Learning SCA Battle (1/15 Teams, $1000 prize)
CHES 2025, GE Wars Organization Committee [link] — *2025*

Winner of First National "Challenger Cup" Quantum Computing Competition ($1500 prize)
Quantum Computing Cloud Platform, Chinese Academy of Sciences — *2022*

National Scholarship ($1000)
Ministry of Education of the People's Republic of China — *2019*

Publications

[LICS '26] Gilles Barthe, Minbo Gao, Jam Kabeer Ali Khan, Matthijs Muis, Ivan Renison, Keiya Sakabe, Michael Walter, **Yingte Xu**, Tianshi Yu, Li Zhou. *Complete Relational Logic for Infinite-Dimensional Quantum Programs with Unbounded Assertions.* [DOI]

[CAV '25] **Yingte Xu**, Li Zhou, and Gilles Barthe. *D-Hammer: Efficient Equational Reasoning for Labelled Dirac Notation.* [arXiv]

[POPL '25] **Yingte Xu**, Gilles Barthe, and Li Zhou. *Automating Equational Proofs in Dirac Notation.* [arXiv]

[TOSEM '25] Yuan Feng, Li Zhou, **Yingte Xu**, and Xiaoquan Xu. *Refinement Calculus of Quantum Programs with Projective Assertions.* [arXiv]

[ASPLOS '23] Yuan Feng and **Yingte Xu**. *Verification of Nondeterministic Quantum Programs.* [arXiv]

Workshop Papers

[PLanQC '26] **Yingte Xu**. *A Unified Assertion-Based Framework for Classical-Quantum Program Verification.* PLanQC 2026, co-located with POPL 2026. [link]

Technical Skills

Programming Languages: Python, C/C++, OCaml, JavaScript, Mathematica, MATLAB
Formal Methods & Proof Automation: Rocq (Coq), Lean 4, term-rewriting engines, SMT and first-order (Vampire) solvers
Machine Learning: PyTorch, Hugging Face (Transformers, Datasets, Accelerate), NumPy, Pandas, Numba
Systems & Tooling: Docker, OpenTelemetry, reactive pipelines (ReactiveX / RxPY), WebSockets, LLM coding-agent workflows

Teaching Experience

Quantum Computing Talent Development Summer School (USTC, online)
Lecturer *2024*
Teaching Assistant *2022–2023*

Linear Algebra (Lanzhou University)
Teaching Assistant *2019*

Academic Service

Subreviewer: ICTAC 2026 (International Colloquium on Theoretical Aspects of Computing)

Artifacts & Projects

Fanal *2025–2026*
Lean 4 project for the foundational formalization and verification of quantum programs. [GitHub]

AI4SMT *2026*
LLM coding-agent framework that optimizes the heuristics of the cvc5 SMT solver against SMT-COMP benchmarks; the resulting solver is registered for SMT-COMP 2026. [GitHub]

D-Hammer *2025*
C++ engine that efficiently decides equations of labelled Dirac notation. Artifact of *D-Hammer* [CAV '25]. [GitHub]

DiracDec *2024–2025*
Mathematica prototype for proof automation of Dirac-notation equations, integrated with its computer-algebra system for reals. Artifact of *Automating Equational Proofs in Dirac Notation* [POPL '25]. [GitHub]

QRefine *2024*
Python prototype IDE for quantum program development with program-refinement support, based on the *Refinement Calculus of Quantum Programs with Projective Assertions* [TOSEM '25]. [GitHub]

NQPV *2023*
Python assistant tool for the formal verification of nondeterministic quantum programs. Artifact of *Verification of Nondeterministic Quantum Programs* [ASPLOS '23]. [GitHub]